Lecture #17: State Classification and Stationary Distributions

Mitchell Scott

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Communication

Definition

We say x *communicates with* y and write $x \to y$ if there is a positive probability of reaching y from x:

$$\rho_{xy} = \mathbb{P}_x(T_y < \infty) > 0.$$

Theorem

If $x \to y$ and $y \to z$, then $x \to z$.

Theorem

If $\rho_{xy} > 0$, *but* $\rho_{yx} < 1$, *then* x *is transient.*

Theorem

If $\rho_{xy} > 0$, but $\rho_{yx} < 1$, then x is transient.

Remark

Stated a different way, if I can get from x to y, but I am not guaranteed to get from y to x, then I could get lost and never return to x.

Theorem

If x is recurrent and $\rho_{xy} > 0$, then $\rho_{yx} = 1$.

Theorem

If x is recurrent and $\rho_{xy} > 0$, then $\rho_{yx} = 1$.

Remark

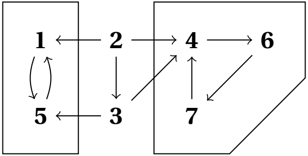
This is the contrapositive of the previous. But also you can think of it as “If I know I return to x infinitely many times, and I can get from x to y, then I have to get to from y back to x.

The Seven State Chain

Example (The Seven State Chain)

Given the below probability transition matrix, determine the type of each state.

	1	**2**	**3**	**4**	**5**	**6**	**7**
1	.7	0	0	0	.3	0	0
2	.1	.2	.3	.4	0	0	0
3	0	0	.5	.3	.2	0	0
4	0	0	0	.5	0	.5	0
5	.6	0	0	0	.4	0	0
6	0	0	0	0	0	.2	.8
7	0	0	0	1	0	0	0


1
2
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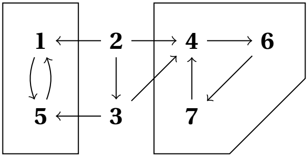
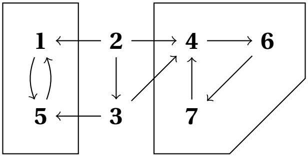


1. The state 2 communicates with 1, which does not communicate with it, so Theorem 1.5 implies that 2 is transient.



1. The state 2 communicates with 1, which does not communicate with it, so Theorem 1.5 implies that 2 is transient.
2. Likewise 3 communicates with 4, which doesn't communicate with it, so 3 is transient.

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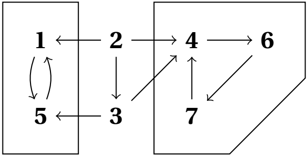
To conclude that all the remaining states are recurrent we will introduce two definitions and a fact.

Definition

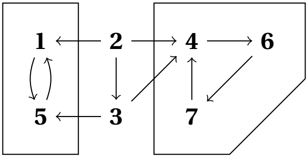
A set A is closed if it is impossible to get out, i.e., if $i \in A$ and $j \notin A$ then $p(i, j) = 0$.



Remark

Definition

Among the closed sets in the last example, some are obviously too big. To rule them out, we need a definition. A set B is called *irreducible* if whenever $i, j \in B$, i communicates with j.



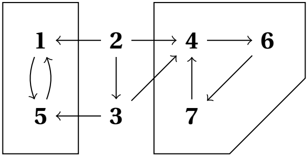
Remark

Theorem

If x is recurrent and $x \to y$, then y is recurrent.

Theorem

In a finite closed set there has to be at least one recurrent state.



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Random Initial Distributions

Question

What happens in a Markov chain when the initial state is random?

Breaking things down according to the value of the initial state and using the definition of conditional probability

If we let $q(i) = \mathbb{P}(X_0 = i)$, then we get

Example (Weather in Ithaca, NY)

Let X_n be the weather on day n in Ithaca, NY, which we assume is either: 1 = rainy, or 2 = sunny. Even though the weather is not exactly a Markov chain, we can propose a Markov chain model for the weather by writing down a transition probability

	1	**2**
1	0.6	0.4
2	0.2	0.8

The table says, for example, the probability a rainy day (state 1) is followed by a sunny day (state 2) is $p(1,2) = 0.4$. A typical question of interest is

Question

What is the long-run fraction of days that are sunny?

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To answer that we ask ourselves that the initial distribution is $q(1) = 0.3$ and $q(2) = 0.7$, meaning 30% of the time it is rainy, and 70% of the time its sunny. In this case

3 State Model (Social Mobility)

Example

Consider the social mobility chain and suppose that the initial distribution: $q(1) = 0.5$, $q(2) = 0.2$, and $q(3) = 0.3$. Multiplying the vector q by the transition probability gives the vector of probabilities at time 1:

Stationary Distribution

Definition

If $qp = q$ then q is called a stationary distribution. If the distribution at time 0 is the same as the distribution at time 1, then by the Markov property it will be the distribution at all times $n \geq 1$.

Question

When does $p^n(x, y) \to \pi(y)$?

Example (Stationary Distribution for 2- state Markov Chain)

Recall the weather chain example.

$$\mathbf{W} = \begin{pmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{pmatrix}$$

We would like to write it in the stationary distribution form.

This means we get the following system of equations:

From high school algebra, we should be able to solve this as there are 2 equations and 2 unknowns. However, we observe that $\pi_1 = 0.5\pi_2$, so plugging into the second equation we get $\pi_2 = \pi_2$, so this is not the way to solve it.
However, we also know that since both equations become

Quickly checking our solution, we arrive at

Now let's try to use this method to a general 2 state Markov chain, where the two states are "1" and "2". Let a be the probability of going from 1 to 2, and b be the probability of going from 2 to 1. Since each row has to sum to 1, this means the general chain looks like

$$\begin{array}{ccc} & \mathbf{1} & \mathbf{2} \\ \mathbf{1} & 1-a & a \\ \mathbf{2} & b & 1-b \end{array}$$

Now to find the stationary distribution, we form the equations

This means we get the following equations:

$$(1-a)\pi_1 + b\pi_2 = \pi_1$$
$$a\pi_1 + (1-b)\pi_2 = \pi_2$$

Lastly, we check that this formula is consistent with the answer that we found for the weather chain problem. There $a = 0.4$, $b = 0.2$, so